

■ PathPilot

2 Core Features + 1 USP — 2-Hour MVP

What PathPilot is

PathPilot is a Skill GPS. It understands where a learner is, where they want to go, and tells them the **next best learning move** instead of overwhelming them with a generic roadmap.

Core loop: **Goal** → **Current Skills** → **Skill Gap** → **Next Skill** → **Free Resource** →
Mini Project → **Proof** → **Adapt**

Core Feature 1 — ■ Skill GPS / Next Best Action

Inspired by Atlas's **Next-Best-Action** and adaptive roadmap concepts, but grounded in PathPilot's deterministic skill graph. Atlas describes next-best-action recommendations as answering the practical question “What should I do next?” rather than giving generic content.

Example:

Learner says	I want to become a Frontend Developer. I know HTML/CSS, have a laptop, and can study 1 hour/day
System finds	HTML ■ CSS ■ JavaScript ■
Skill gap	JavaScript Fundamentals
Next Move	Learn JavaScript Fundamentals
Then	Free resource + small project + reason why this step comes next

Important architecture: the Skill Engine determines prerequisite truth; Gemini explains and personalizes the recommendation. Gemini does not invent prerequisites or resource URLs.

Core Feature 2 — ■■ Learn → Build → Prove → Adapt

This compresses Atlas's progress/adaptation ideas and KAL-AI's learner-memory/personalized guidance into one practical checkpoint that is achievable in the MVP.

Example checkpoint: JavaScript Functions

LEARN	Use a curated free resource
BUILD	Build a small calculator
PROVE	Acceptance criteria: functions, input, operations, basic error handling
ADAPT	Completed → next skill. Stuck → ask why and give a simpler explanation/re-plan.

■ USP — Proof-Based Next Move

“PathPilot doesn't just tell you what to learn. It tells you what to do next — and asks you to prove the skill before moving forward.”

This differentiates PathPilot from a generic AI roadmap generator. The learner is not simply consuming content; each important step ends with a small, verifiable demonstration.

The Product Loop

Goal → Current State → Skill Gap → ■ Next Best Move → Free Resource → Mini Project → Proof → Ready / Stuck → Re-plan → Next Best Move

What we borrow from Atlas + KAL-AI

Source	Use in PathPilot	MVP decision
Atlas	Next-Best-Action	YES — core feature
Atlas	Adaptive roadmap	YES — only completed/stuck branching
Atlas	Learner state / memory	YES — tiny saved learner state
KAL-AI	Grounded/approved content	YES — curated resources
KAL-AI	Personalized guidance	YES — Gemini explanation
KAL-AI	Video/PDF/audio understanding	NO — too much for 2 hours
Atlas + KAL	Analytics, behavior tracking, communities, rich dashboards	NO — future scope

■■ 2-Hour MVP — Build Only This

1. Quick intake: goal, current skills, time/day, device, constraints.
2. Skill graph + gap analysis for one or two demo goals.
3. “Your Next Move” screen with one free resource and one project.
4. Skill checkpoint with Completed / I'm Stuck.
5. Gemini for explanation/personalization and simple re-planning.

Do NOT build now: voice assistant, AI avatar, job scraper, blockchain passport, full mentor dashboard, community, automated GitHub code review, complex analytics, or a huge resource database.

Hackathon one-liner

“Don't follow a roadmap. Follow your next move. PathPilot converts learning into a closed loop: Identify → Learn → Build → Prove → Adapt.”